

*Notebook on*  
**Peripheral & Interfacing**

*Prepared by*  
**Shahriar Arefin**

**ID: 22101040**

**Section: (A)**

*Faculty*  
**Atia Rahman Orthi**

**Semester: Fall 2024**

# Contents

|          |                                           |           |
|----------|-------------------------------------------|-----------|
| <b>1</b> | <b>Sensors and Modules</b>                | <b>1</b>  |
| 1.1      | Sensors . . . . .                         | 1         |
| 1.1.1    | DHT11/DHT22 Sensor . . . . .              | 1         |
| 1.1.2    | BMP180 Barometric Sensor . . . . .        | 1         |
| 1.1.3    | Rain Sensor . . . . .                     | 2         |
| 1.1.4    | Soil Moisture Sensor . . . . .            | 2         |
| 1.1.5    | Gas/Smoke Sensor . . . . .                | 3         |
| 1.1.6    | Ultrasonic Sensor . . . . .               | 3         |
| 1.1.7    | PIR Motion Sensor . . . . .               | 4         |
| 1.1.8    | RFID . . . . .                            | 4         |
| 1.1.9    | Microprocessor Sound Sensor . . . . .     | 5         |
| 1.1.10   | Relay Module . . . . .                    | 5         |
| 1.1.11   | L298N Motor Driver . . . . .              | 5         |
| 1.1.12   | Servo Motor . . . . .                     | 6         |
| 1.1.13   | Bluetooth Sensor . . . . .                | 6         |
| 1.1.14   | IR Sensor . . . . .                       | 6         |
| 1.1.15   | Water Level Sensor . . . . .              | 6         |
| 1.1.16   | Arduino Uno . . . . .                     | 7         |
| <b>2</b> | <b>IoT Projects</b>                       | <b>8</b>  |
| 2.1      | Some IoT Projects Examples . . . . .      | 8         |
| 2.1.1    | Smart Agriculture . . . . .               | 9         |
| 2.1.2    | 2WD and 4WD Robot . . . . .               | 11        |
| <b>3</b> | <b>3D Printer</b>                         | <b>12</b> |
| 3.1      | Binder Jetting . . . . .                  | 12        |
| 3.2      | Material Jetting . . . . .                | 12        |
| 3.3      | Stereolithography . . . . .               | 13        |
| 3.4      | Fused Deposition Modeling (FDM) . . . . . | 13        |

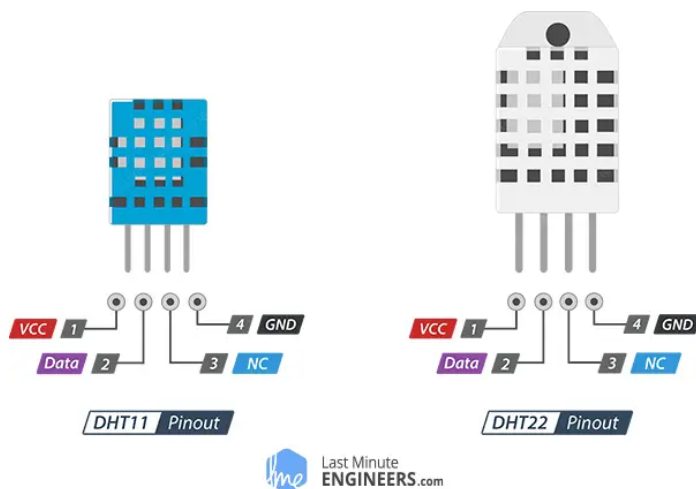
# Chapter 1

## Sensors and Modules

### 1.1 Sensors

#### 1.1.1 DHT11/DHT22 Sensor

DHT11/DHT22 sensor is used to measure the temperature and humidity. These sensors contain a chip that does analog to digital conversion and spits out a digital signal with the temperature and humidity.

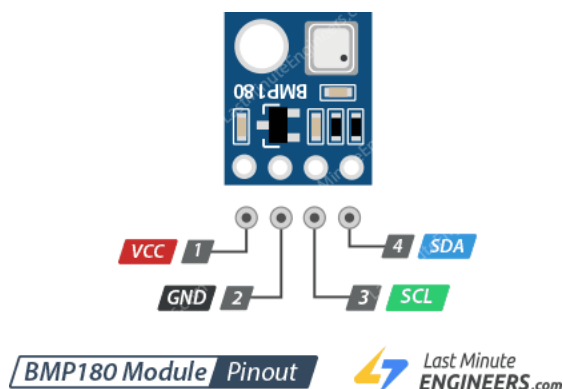


#### Logical Code

```
dht.readHumidity()  
dht.readTemperature()  
dht.readTemperature(true)
```

#### 1.1.2 BMP180 Barometric Sensor

BMP180 Barometric Sensor measures the absolute pressure of the air around it. It has a measuring range from 300hPa to 1100hPa with an accuracy down to 0.02hPa. It can also measure altitude and temperature.

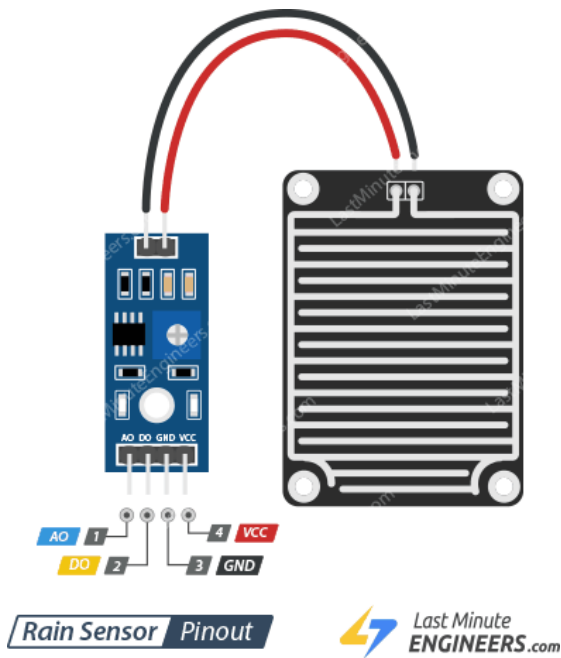


#### Logical Code

```
bmp.readTemperature()  
bmp.readPressure()  
bmp.readAltitude()
```

### 1.1.3 Rain Sensor

The rain sensor is used to detect water and it can detect beyond what a humidity sensor can. **Wet:** the resistance increases, and the output voltage decreases and **Dry:** the resistance is lower, and the output voltage is higher.

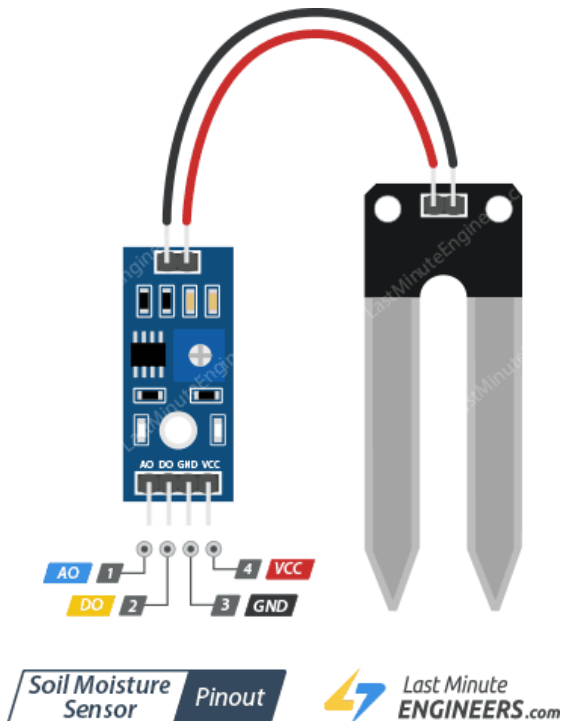


#### Logical Code

```
digitalRead(rainPin) → int  
analogRead(rainPin) → int
```

### 1.1.4 Soil Moisture Sensor

The soil moisture sensor or the hygrometer is usually used to detect the humidity of the soil. So, it is perfect to build an automatic watering system or to monitor the soil moisture of your plants.



#### Logical Code

```
digitalRead(PIN) → int  
analogRead(PIN) → int
```

### 1.1.5 Gas/Smoke Sensor

The MQ-2 smoke sensor is sensitive to smoke and to the following flammable gases: LPG, butane, propane, methane, alcohol and hydrogen. The resistance across the sensor is different depending on the type of the gas. The greater the gas concentration, the greater the output voltage.

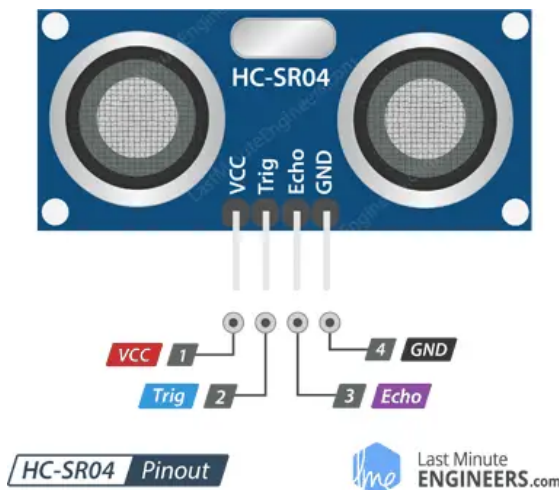


#### Logical Code

```
digitalRead(PIN) → int
analogRead(PIN) → int
```

### 1.1.6 Ultrasonic Sensor

The HC-SR04 ultrasonic sensor uses sonar to determine distance. It offers excellent non-contact range detection with high accuracy and stable readings in an easy-to-use package. From 2 cm to 400 cm or 1" to 13 feet.

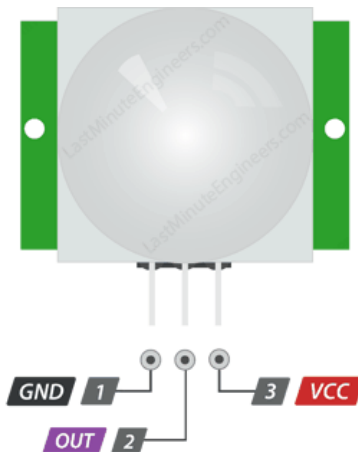


#### Logical Code

```
digitalWrite(trigPin, LOW)
delayMicroseconds(2)
digitalWrite(trigPin, HIGH)
delayMicroseconds(10)
digitalWrite(trigPin, LOW)
long duration = pulseIn(echoPin, HIGH)
cm = (duration/2) / 29.1
inches = (duration/2) / 74
```

### 1.1.7 PIR Motion Sensor

The PIR motion sensor is ideal to detect movement. PIR stand for “Passive Infrared”. Basically, the PIR motion sensor measures infrared light from objects in its field of view. So, it can detect motion based on changes in infrared light in the environment.



PIR Sensor Pinout



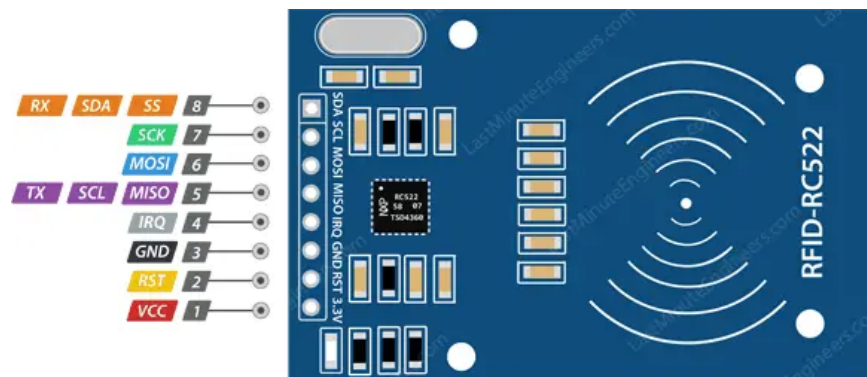
#### Logical Code

`digitalRead(PIN) → HIGH/LOW`

\*\* Check **state change** using **if-else** & **save** the current state

### 1.1.8 RFID

RFID means radio-frequency identification. RFID uses electromagnetic fields to transfer data over short distances. Each tag has its own unique identification (UID). Two-way radio transmitter-receiver, the reader, that send a signal to the tag and read its response.



RC522 Pinout

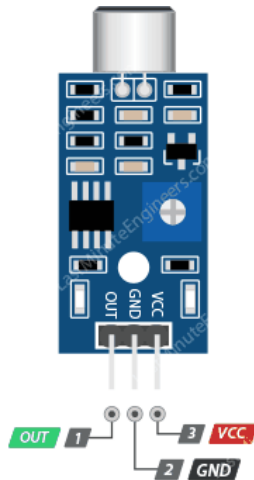


```
byte authorizedUID[] = {0xAB, 0xCD, 0xEF, 0x12};
```

```
bool checkAccess() {
    if (!mfr522.PICC_IsNewCardPresent() || !mfr522.PICC_ReadCardSerial()) {
        return false;
    }
    for (byte i = 0; i < mfr522.uid.size; i++) {
        if (mfr522.uid.uidByte[i] != authorizedUID[i]) {
            return false;
        }
    }
    return true;
}
```

### 1.1.9 Microprocessor Sound Sensor

Microprocessor sound sensor gives a measurement of how loud a sound is.



#### Logical Code

```
digitalRead(PIN) → int
```

Sound Sensor Pinout



### 1.1.10 Relay Module

A relay is an electrically operated switch of mains voltage. It can be turned on or off, letting the current go through or not.



#### Logical Code

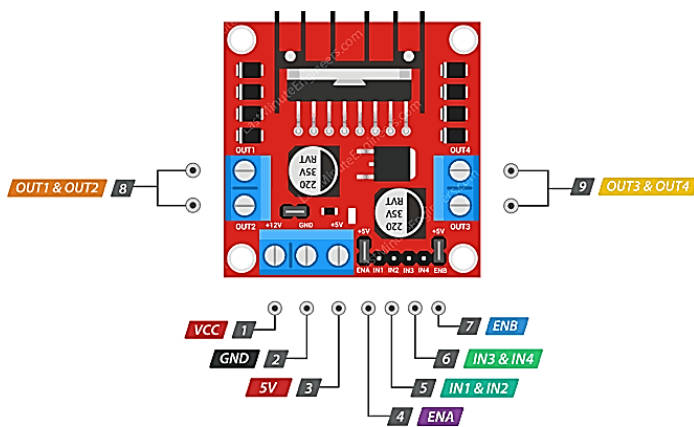
```
digitalWrite(relayPIN, HIGH/LOW)
```

Relay Module Pinout



### 1.1.11 L298N Motor Driver

The L298N 4WD Motor Driver is a versatile and robust H-bridge driver module designed for controlling DC motors with ease and precision.



#### Logical Code

```
digitalWrite(in1, HIGH) → direction
```

```
digitalWrite(in2, LOW) → direction
```

```
analogWrite(enA, 0-255) → speed
```

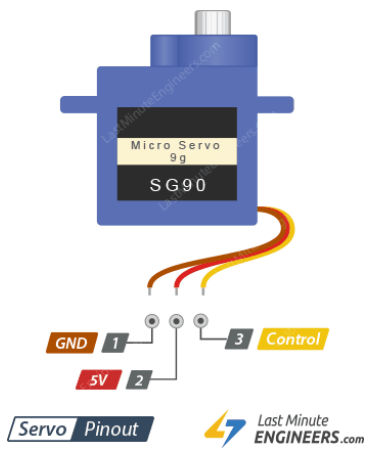
HIGH + LOW = Forward

LOW + HIGH = Backward

L298N Module Pinout

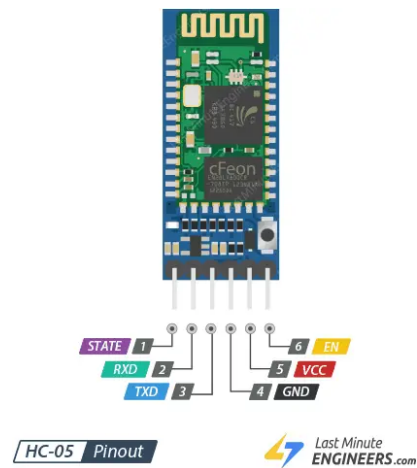


### 1.1.12 Servo Motor



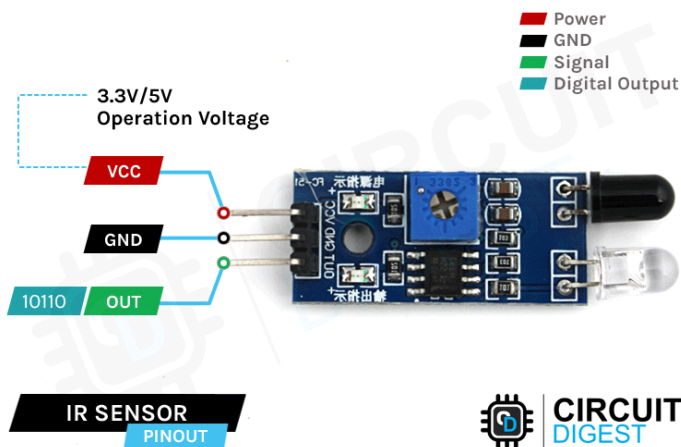
```
myServo.write(0-180)
```

### 1.1.13 Bluetooth Sensor



```
bluetoothSerial.available() | read() | write()
```

### 1.1.14 IR Sensor

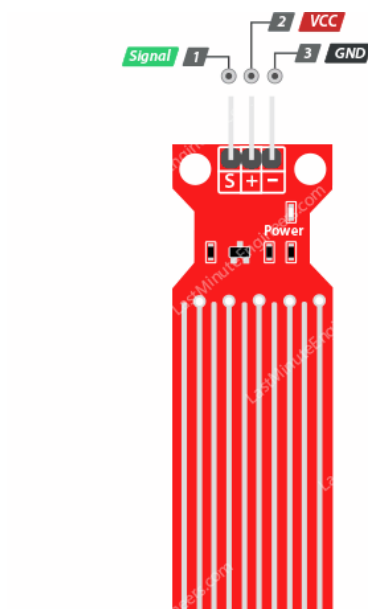


#### Logical Code

```
int irValue = digitalRead(irPin)

if (irValue == HIGH){
  Serial.println("No Obstacle Detected");
} else {
  Serial.println("Obstacle Detected");
}
```

### 1.1.15 Water Level Sensor



#### Logical Code

```
int waterLevel = analogRead(PIN)
```

Outputs: 0 to 1023

Higher values indicate more water.





# Chapter 2

## IoT Projects

### 2.1 Some IoT Projects Examples

#### Smart Poultry and Fish Farming System

1. NodeMCU (ESP8266)
2. LCD Display
3. DHT11/DHT22
4. Relay Module
5. SMPS
6. Ultrasonic Sensor
7. Moisture Sensor
8. pH Sensor
9. TDS Sensor
10. Turbidity Sensor
11. MQ-135 Gas Sensor
12. Water Pump
13. Light Dependent Resistor (LDR)
14. Servo Motor

#### Obstacle Avoider Robot

1. Arduino UNO / ESP8266
2. Motor Driver
3. Servo Motor
4. Ultrasonic sensor
5. Gear Motor
6. 18650 Battery

#### Line Follower Robot

1. Arduino UNO / ESP8266
2. Gear Motor
3. Motor Driver
4. IR sensor
5. 18650 Battery

#### Smart Plant Monitoring System

1. Arduino UNO / ESP8266
2. PIR Motion sensor
3. DHT11 Temperature sensor
4. Soil Moisture sensor
5. Relay module
6. 18650 Battery

#### Metal Detector Robot

1. Arduino UNO / ESP8266
2. Gear Motor
3. Motor Driver
4. Metal Detector
5. LCD display
6. Joystick Module
7. 18650 Battery

#### Voice Controlled Car

1. Arduino UNO / ESP8266
2. Gear Motor
3. Servo Motor
4. Motor Driver
5. Ultrasonic sensor
6. Bluetooth sensor
7. 18650 Battery

### 2.1.1 Smart Agriculture

#### Components:

- |                                   |   |                                                       |
|-----------------------------------|---|-------------------------------------------------------|
| 1. ESP8266                        | : | Micro-controller for IoT connectivity.                |
| 2. PIR Motion sensor              | : | To detect motion in the farm.                         |
| 3. DHT11 Temperature sensor       | : | To monitor temperature and humidity.                  |
| 4. Soil Moisture sensor           | : | To measure soil moisture levels.                      |
| 5. Relay module                   | : | To control devices like pumps or lights.              |
| 6. 18650 Battery                  | : | Power supply for the system.                          |
| 7. LDR (Light Dependent Resistor) | : | To monitor light intensity for plant growth.          |
| 8. Water Pump                     | : | To irrigate plants based on soil moisture levels.     |
| 9. pH Sensor                      | : | To monitor soil pH for optimal plant growth.          |
| 10. OLED Display                  | : | To display sensor readings and system status locally. |

#### Pseudo Code:

```

BEGIN
  INITIALIZE the sensors and actuators

  LOOP FOREVER:
    === Sensor Readings ===
    READ temperature, humidity from DHT11
    READ soilMoistureValue from Soil Moisture Sensor
    READ pHValue from pH Sensor
    READ lightLevel from LDR
    READ motionDetected from PIR Sensor (HIGH/LOW)

    === Irrigation Logic ===
    IF soilMoistureValue < THRESHOLD:
      TURN ON Relay (activate water pump)
      WAIT for irrigationTime seconds
      TURN OFF Relay
    ELSE:
      KEEP water pump OFF

    === Motion Detection Logic ===
    IF motionDetected is HIGH:
      LOG or DISPLAY "Motion detected"
      Send alert to user

    === Display Logic ===
    CLEAR OLED screen
    DISPLAY(temperature,humidity,soil_moisture,pH)

    DELAY 1000
  END LOOP
END

```

**Iteration:**

```
void loop() {  
  // Sensor Readings  
  float temperature = dht.readTemperature();  
  float humidity = dht.readHumidity();  
  int soilMoistureValue = analogRead(soilMoisturePin);  
  float pHValue = readPH();  
  int lightLevel = analogRead(ldrPin);  
  bool motionDetected = digitalRead(pirPin) == HIGH ? true : false;  
  
  // Irrigation Logic  
  if (soilMoistureValue < moistureThreshold) {  
    digitalWrite(WATER_PUMP_RELAY, HIGH);  
    delay(irrigationTime * 1000);  
    digitalWrite(WATER_PUMP_RELAY, LOW);  
  } else {  
    digitalWrite(WATER_PUMP_RELAY, LOW);  
  }  
  
  // Motion Detection Logic  
  if (motionDetected) {  
    Serial.println("Motion detected!");  
  }  
  
  // Display Logic  
  oled.clearDisplay();  
  oled.setCursor(0, 0);  
  oled.print("Temp: "); oled.println(temperature);  
  oled.print("Humidity: "); oled.println(humidity);  
  oled.print("Soil: "); oled.println(soilMoistureValue);  
  oled.print("pH: "); oled.println(pHValue);  
  oled.print("Light: "); oled.println(lightLevel);  
  oled.print("Motion: "); oled.println(motionDetected ? "Yes" : "No");  
  oled.display();  
  
  delay(1000);  
}
```

## 2.1.2 2WD and 4WD Robot

### Pseudo Code:

```

LOOP FOREVER:
  READ leftSensor and rightSensor

  IF leftSensor IS WHITE AND rightSensor IS WHITE THEN
    MOVE forward
  ELSE IF leftSensor IS BLACK AND rightSensor IS WHITE THEN
    TURN right
  ELSE IF leftSensor IS WHITE AND rightSensor IS BLACK THEN
    TURN left
  ELSE
    STOP motors
END LOOP

```

Black lines reflect less IR.

So, IR sensors return LOW when over black and HIGH when over white.

HIGH + LOW = Forward

LOW + HIGH = Backward

### Iteration:

```

void loop() {
  int RS = digitalRead(IR_RIGHT_PIN);
  int LS = digitalRead(IR_LEFT_PIN);

  // Both sensors see white => Move forward
  if (LS == 1 && RS == 1) {
    setMotors(HIGH, HIGH);
  }

  // Right sensor sees black, turn left
  if (LS == 1 && RS == 0) {
    setMotors(LOW, HIGH);
  }

  // Left sensor sees black, turn right
  if (LS == 0 && RS == 1) {
    setMotors(HIGH, LOW);
  }

  // Both sensors see black, stop
  if (LS == 0 && RS == 0) {
    stopMotors();
  }
}

// Combined control function
const setMotors(bool isLeftForward, bool isRightForward) {
  digitalWrite(in1, isLeftForward); digitalWrite(in2, !isLeftForward);
  digitalWrite(in3, isRightForward); digitalWrite(in4, !isRightForward);
  analogWrite(enA, MAX_SPEED); analogWrite(enB, MAX_SPEED);
}

const stopMotors() {
  analogWrite(enA, 0);
  analogWrite(enB, 0);
}

```

# Chapter 3

## 3D Printer

### 3.1 Binder Jetting

In Binder Jetting, A roller or blade spreads a fresh, even layer of part material powder over the build surface. The print head **selectively sprays the binder** onto specific regions of the powder bed, as defined by the 3D model. The **binder fuses the powder particles together** to form solid layers of the desired part geometry. The build **platform lowers incrementally**, and the process repeats until the part is complete.

#### Post-Processing:

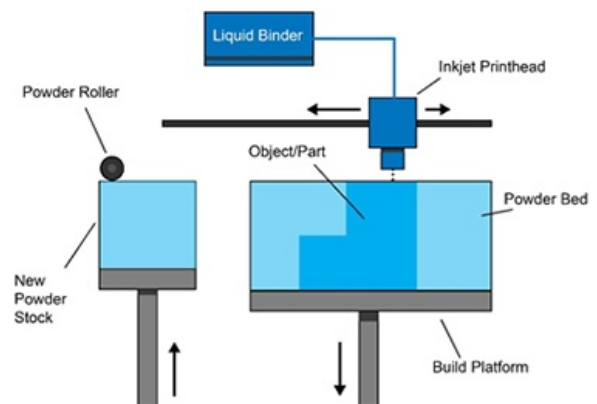
- **Depowdering:** to remove the excess powder.
- **Sintering & Infiltration:** to improve appearance and durability.

#### Key Features:

- **No supports needed** (powder supports overhangs).
- Works with metals, ceramics, sand, and polymers.
- Faster and more **cost-effective** than laser-based powder-bed methods.

#### Limitations:

- Lower mechanical strength compared to laser based AM.
- Rougher surface finish compared to other methods.
- Require additional post-processing for strength and finish.



### 3.2 Material Jetting

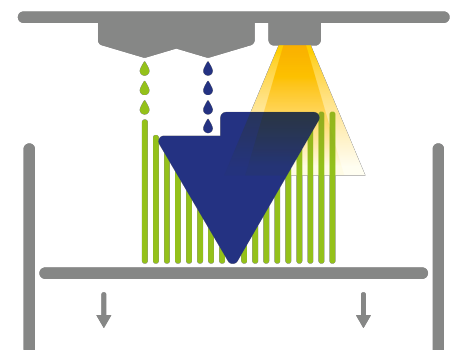
In Material Jetting, droplets of actual build materials typically liquid photopolymers or resins are deposited directly onto the build platform through multiple print heads. Each deposited layer is then instantly cured (hardened) using a UV light, ensuring dimensional stability. Support materials are jetted alongside and later removed once finished.

#### Key Features:

- Produces highly accurate parts with excellent smoothness.
- Allows multi-material printing of different properties in a single part.
- Simultaneous deposition of multiple materials reduces print time.

#### Limitations:

- Primarily limited to photopolymer resins which is quite expensive.
- Material properties degrade under UV exposure over time.
- Support removal can be tedious for intricate designs.



### 3.3 Stereolithography

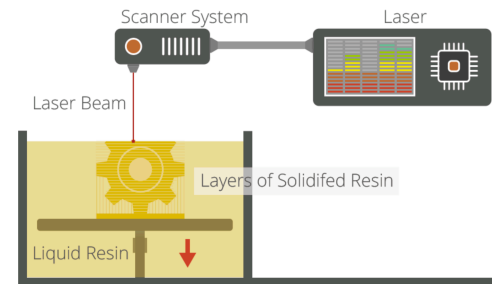
Stereolithography (SL) is a laser-based 3D printing process that uses photopolymer resin and cure to form a solid in a very precise way to produce very accurate parts. The photopolymer resin is held in a vat with a movable platform inside. A laser beam is directed in the X-Y axes across the surface of the resin according to the 3D data supplied to the machine (the .stl file), whereby the resin hardens precisely where the laser hits the surface.

#### Key Features:

- Excellent dimensional precision and resolution.
- Smooth surface finish and ideal for visual prototypes: dental, jewelry.

#### Limitations:

- Photopolymers degrade or brittle under UV exposure over time.
- Vat size restricts part dimensions and resins are expensive.
- Support removal can be tedious for intricate designs.



### 3.4 Fused Deposition Modeling (FDM)

The thermoplastic filament is fed into a heated print head, which moves in the X and Y axes, laying down the melted material according to the 3D model data supplied to the machine. Once a layer is complete, the build platform lowers slightly along the Z-axis, and the next layer is deposited directly on top of the previous one. Each layer hardens as it is deposited and bonds to the previous layer.

#### Key Features:

- Uses materials like PLA, ABS, Nylon; suitable for both prototyping and functional parts.
- Can use breakaway or water-soluble support materials for complex geometries.
- One of the most affordable and accessible 3D printing methods.

#### Limitations:

- The process can be slow for some part geometries.
- Layer-to-layer adhesion can be a problem, resulting in parts that are not watertight.
- Requires post-processing using Acetone to resolve the issue.

